

ICT80004 Weekly Communication – Week #09

Student Name: Arun Ragavendhar Arunachalam Palaniyappan ID: 104837257

Organisation: Commonwealth Scientific and Industrial Research Organisation (CSIRO)

Industry Supervisor: Dr. Shigang Liu

Date Prepared: 03/10/2025 Internship Week #: 9

Day	Date	Task(s) completed
1	Monday 30 Sep 2025 8 hours	- Conducted a full review of the paper <i>“Trust Me, I Know This Function: Hijacking LLM Static Analysis using Bias”</i> and extracted key mechanisms (abstraction bias, Familiar Pattern Attacks). - Implemented and tested the complete FPA experiment pipeline: - fpa_demo.py (vowel-check deception) - fpa_generator_skeleton.py (black-box generator scaffold) - fpa_generator_complete.py and fpa_generator_complete_fixed.py (full pipeline with Mock LLM + OpenAI API modes). - Executed mock experiments with vowel membership and nth_prime off-by-one cases, producing CSV outputs confirming divergence between runtime behavior and LLM predictions. - Prepared detailed slide decks explaining the TL;DR, threat model, formal conditions, and case study walkthrough (nth_prime). - Documented defensive interpretations (anti-plagiarism, anti-scraping) from the paper and mapped them to our use cases.
2	Wednesday 1 Oct 2025 8 hours	- Confirmed the paper’s claim: tiny edits (P’) reliably mislead models while preserving runtime correctness. - Attack success demonstrated in both vowel membership (drop_u) and numeric (nth_prime) examples under the mock LLM setting. - Abstraction bias is structural: the model “assumes” archetypal behavior rather than evaluating details. - Prompt-based warnings are insufficient; multiple layers of checks will be necessary. - FPA’s show clear potential for both offensive evasion and defensive watermarking/obfuscation.

Total hours completed for the week: 16

Plans for next week: #09 week (6 Oct – 10 Oct 2025)

- Expand the FPA case library beyond vowels: LSWR comparator tweaks, modulo normalization, min/max boundary swaps.
- Translate confirmed successful patterns into C, Rust, and Go to validate cross-language transferability.
- Run robust prompt and reasoning model baselines for comparative evaluation.
- Generate aggregate CSV datasets and plots showing success rates and transfer across cases.
- Begin drafting a structured FPA casebook with runtime vs. LLM predictions, code, and prompts for red/blue-team benchmarking.

Screenshot of Timely EMAIL communication update to the Supervisor at the end of week #07 sent on 3rd October 2025 Friday at 3:48 PM (15:48).

Weekly Communication / Reflection update - #09 week, 29 Sep - 03 Oct 2025

A

Arun Ragavendhar

<arunragavendhar.1999@gmail.com>

to Shigang

15:48 (4 minutes ago)

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Dear Dr. Liu,

I hope you are well. Please find my Week 9 update (29 Sept – 3 Oct 2025) below.

Activities completed this week (Total: 17 hours)

- Conducted a full review of the paper “*Trust Me, I Know This Function: Hijacking LLM Static Analysis using Bias*” and extracted key mechanisms (abstraction bias, Familiar Pattern Attacks).
- Implemented and tested the complete FPA experiment pipeline:
 - `fpa_demo.py` (vowel-check deception)
 - `fpa_generator_skeleton.py` (black-box generator scaffold)
 - `fpa_generator_complete.py` and `fpa_generator_complete_fixed.py` (full pipeline with Mock LLM + OpenAI API modes).
- Executed mock experiments with vowel membership and `nth_prime` off-by-one cases, producing CSV outputs confirming divergence between runtime behavior and LLM predictions.
- Prepared detailed slide decks explaining the TL;DR, threat model, formal conditions, and case study walkthrough (`nth_prime`).
- Documented defensive interpretations (anti-plagiarism, anti-scraping) from the paper and mapped them to our use cases.

Key findings / interim observations

- Confirmed the paper’s claim: tiny edits (*P*) reliably mislead models while preserving runtime correctness.
- Attack success demonstrated in both vowel membership (`drop_u`) and numeric (`nth_prime`) examples under the mock LLM setting.
- Attack success demonstrated in both vowel membership (`drop_u`) and numeric (`nth_prime`) examples under the mock LLM setting.
- Abstraction bias is structural: the model “assumes” archetypal behavior rather than evaluating details.
- Prompt-based warnings are insufficient; multiple layers of checks will be necessary.
- FPAs show clear potential for both offensive evasion and defensive watermarking/obfuscation.

Plan for next week (Week 10: 6–10 Oct 2025)

- Expand the FPA case library beyond vowels: LSWR comparator tweaks, modulo normalization, min/max boundary swaps.
- Translate confirmed successful patterns into C, Rust, and Go to validate cross-language transferability.
- Run robust prompt and reasoning model baselines for comparative evaluation.
- Generate aggregate CSV datasets and plots showing success rates and transfer across cases.
- Begin drafting a structured FPA casebook with runtime vs. LLM predictions, code, and prompts for red/blue-team benchmarking.

Kind regards,
Arun Ragavendhar A. P.
ICT80004 Internship Student – CSIRO Data61

↩ Reply

➡ Forward

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